

JESSE AMOAKO ASIMENG

Computational Biologist & Data Scientist

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PROFESSIONAL SUMMARY

Computational Biologist with 5+ years of experience at the intersection of data science, machine learning, and bioinformatics. Currently pursuing a PhD at the German Centre for Neurodegenerative Diseases (DZNE), developing graph-based deep learning models and federated learning pipelines for precision medicine. Proven track record of building production-quality R packages, deploying interactive dashboards, and translating complex omics datasets into actionable biological insights. Recognized as the top master's graduate in the Faculty of Medicine and Health Sciences at Stellenbosch University in 2023.

CORE COMPETENCIES

- Machine Learning & Deep Learning
- Federated / Swarm Learning
- Graph Neural Networks (PyTorch)
- Interactive Dashboards (Shiny, Streamlit)
- Single-Cell & Bulk RNA-seq Analysis
- Statistical Modeling & Inference
- Reproducible Workflows (Docker, Nextflow)
- Scientific Communication & Mentoring

PROFESSIONAL EXPERIENCE

Doctoral Researcher

Mar 2024 – Present

German Centre for Neurodegenerative Diseases (DZNE) – Bonn, Germany

- Engineered containerized bioinformatics workflows using Docker and Singularity, reducing environment setup time and ensuring full reproducibility across HPC clusters.
- Designed and trained graph attention network (scGAT) models on single-cell RNA-seq data, achieving robust disease classification performance.
- Implemented federated learning pipelines (Flower framework) enabling decentralized model training across 6 German university hospitals while preserving patient data privacy.
- Built interactive data visualization dashboards (R Shiny, Streamlit) used by 10+ scientific collaborators for real-time data exploration and hypothesis generation.
- Supervised 2 master's students on data analysis and software development projects, guiding them from problem formulation to thesis completion.
- Collaborated with clinicians, immunologists, and statisticians to translate research questions into scalable computational solutions.

Data Scientist Intern

Jun 2023 – Feb 2024

Soldersmith Intelligence – Cape Town, South Africa

- Researched, developed, and optimized computer vision algorithms, improving model accuracy for client-facing AI products.
- Contributed to full-stack web development (Django) to enhance product features and streamline internal data workflows.

Writing Consultant

Feb 2023 – Feb 2024

Stellenbosch University Writing Lab – Stellenbosch, South Africa

- Provided one-on-one academic writing coaching to graduate and undergraduate students, strengthening scientific communication and argumentation skills.

Research Assistant

Sep 2018 – Jun 2021

Kumasi Centre for Collaborative Research in Tropical Medicine – Kumasi, Ghana

- Performed molecular diagnostics (DNA extraction, conventional and real-time PCR) to identify meningococcal and influenza pathogens across hundreds of clinical samples.

- Co-developed standard operating procedures (SOPs) for the multi-site “Febrile Illness in Sub-Saharan Africa” study.
- Conducted systematic literature reviews and supported clinical data collection via structured questionnaires.

EDUCATION

PhD, Computational Life Sciences <i>University of Bonn</i> Thesis: Swarm Learning and Single-Cell Omics: Advancing Global Precision Medicine in Infectious Diseases	Mar 2024 – Present
MSc, Molecular Biology (Bioinformatics) <i>Stellenbosch University</i> Cum Laude (89/100) Thesis: Pipeline and Tools for the Analysis of Multiplexed ELISA Data	Mar 2021 – Mar 2023
BSc, Biological Sciences <i>Kwame Nkrumah University of Science and Technology</i> Second Class Honours (Upper Division)	Sep 2014 – Jun 2018

KEY PROJECTS

LuminexR – Open-Source R Package github.com/Asimeng/LuminexPipeline • Led end-to-end development of an R package for reproducible processing and analysis of Luminex (Multiplex ELISA) data, including raw file import, data cleaning, statistical summaries, and missing-value management.
Graph-Based Deep Learning for Disease Classification • Training a graph attention network on scRNA-seq data to classify disease states and identify key biomarkers; deploying the model in a federated learning setting across multiple hospital sites.

HONOURS & AWARDS

Travel Grant – Pandemic Sciences Institute, University of Oxford £1,100 grant to attend the International Pandemic Sciences Conference (Jun–Jul 2025).	May 2025
Novartis Next Generation Scientist Fellowship Competitively awarded fellowship recognizing emerging research talent (declined due to relocation-based eligibility change).	Mar 2024
Rector’s Award for Excellent Achievement – Stellenbosch University Top master’s student in the Faculty of Medicine and Health Sciences.	Oct 2023
DAAD In-Region Scholarship Full scholarship for MSc studies in Molecular Biology (Bioinformatics), funded Jul 2021–Jun 2023.	Jul 2021

SELECTED PRESENTATIONS

- **Oral:** “Swarm Learning for Precision Medicine and Pandemic Preparedness” – International Pandemic Sciences Conference, Oxford, UK (Jun 2025).
- **Oral:** “LuminexR: a Tool for the Robust and Reproducible Analysis of Multiplexed ELISA Data” – EARTH Summer School Conference, Hamburg, Germany (Aug 2023).
- **Poster:** “Decentralized Graph Learning Across Heterogeneous Single-Cell Transcriptomics Data Sets” – Advances in Single-Cell Technologies, Freiburg, Germany (Dec 2025).
- **Poster:** “Pipeline and Tools for the Robust and Reproducible Analysis of Multiplexed ELISA Data” – BIO2022, Stellenbosch, South Africa (Apr 2022).

TECHNICAL SKILLS

Programming	Python, R, SQL, Django, Git
ML / AI Frameworks	PyTorch, PyTorch Geometric, scikit-learn, Flower (Federated Learning), Explainable AI
Data Science Workflow	Nextflow, Docker, Singularity, HPC

Statistical Methods	GLMMs, Bayesian Additive Regression Trees, Hypothesis Testing
Visualization	ggplot2, R Shiny, Streamlit, Matplotlib
Domain Expertise	Single-Cell & Bulk RNA-seq, Multiplex ELISA, Clinical Data Analysis

LANGUAGES

English (Proficient – C2) | **German** (Intermediate – B1) | **Twí** (Native)